Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A boating club consists of 82 members, each member is either a sailboat owner or a powerboat owner. If 53 members owned sailboats and 38 members owned powerboats, the number of members owned both sailboat and powerboat is

(A) 6

(B) 7

(C) 9

(D) 4

Q.2

The relation R defined by '>' on the set $\mathbf{N}$ is

(A) reflexive

(B) symmetric

(C) transitive

(D) equivalence relation

Q.3

A, B, C are the sets of letters needed to spell the words STUDENT, PROGRESS and CONGRUENT, respectively. Then $n(A \cup (B \cap C))$ is equal to

(A) 8

(B) 9

(C) 10

(D) 11

Q.4

Let $X = \{x : x = n^3 + 2n + 1, n \in \mathbf{N}\}$ and $Y = \{x : x = 3n^2 + 7, n \in \mathbf{N}\}$ then

(A) $X \cap Y$ is a subset of $\{x : x = 3n + 5, n \in \mathbf{N}\}$

(B) $X \cap Y \subseteq \{x : x = n^2 + n + 1, n \in \mathbf{N}\}$

(C) $34 \in X \cap Y$

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(D) none of these

Q.5

The relation aRb defined by a is factor of b on $\mathbf{N}$ is not

(A) reflexive

(B) transitive

(C) anti symmetric

(D) symmetric

Q.6

If A, B, C are three non-empty sets such that $A \cap B = \phi, B \cap C = \phi$, then

(A) $A = C$

(B) $A \subset C$

(C) $C \subset A$

(D) none of these

Q.7

Two finite sets have m and n elements respectively. The total number of subsets of first set is 56 more than the total number of subsets of the second set. The values of m and n respectively are

(A) 7,6

(B) 6,3

(C) 5,1

(D) 8,7

Q.8

Let W denote the words in the English Dictionary. Define the relation $\mathbf{R}$ by $\mathbf{R} = \{(x, y) \in W \times W :$ the words x and y have at least one letter common$\}$, then $\mathbf{R}$ is

(A) reflexive, not symmetric and transitive

(B) not reflexive, symmetric and transitive

(C) reflexive, symmetric and not transitive

(D) reflexive, symmetric and transitive

Q.9

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Let $R = \{(x, y) : x, y \in A, x + y = 5\}$, where $A = \{1, 2, 3, 4, 5\}$ then

(A) R is not reflexive, symmetric and not transitive

(B) R is an equivalence relation

(C) R is reflexive, symmetric but not transitive

(D) R is not reflexive, not symmetric but transitive

Q.10

In a factory 70% of the workers like oranges and 64% like apples. If $x\%$ like both oranges and apples, then

(A) $x \leq 34$

(B) $x \geq 64$

(C) $34 \leq x \leq 64$

(D) none of these.

Q.11

If $A = \{4^n - 3n - 1 \mid n \in \mathbf{N}\}$ and $B = \{9n - 9 : n \in \mathbf{N}\}$, then $A \cup B$ is equal to

(A) B

(B) A

(C) $\mathbf{N}$

(D) none of these

Q.12

Let U be a Universal set and $n(U) = 12$. If $A, B \subseteq U$ are such that $n(B) = 6$ and $n(A \cap B) = 2$ then $n\left(A \cup B'\right)$ is equal to

(A) 6

(B) 10

(C) 7

(D) 8

Q.13

Let R be a relation on $\mathbf{R}$ defined as aRb if $|a| \leq b$ Then, relation R is

(A) reflexive

(B) symmetric

(C) transitive

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(D) not antisymetric

Q.14

The relation R defined on the set $A = \{1, 2, 3, 4, 5\}$ by $R = \{(x, y) : |x^2 - y^2| < 16\}$ is given by

(A) $R_1 = \{(1, 1), (2, 1), (3, 1), (4, 1), (2, 3)\}$

(B) $R_2 = \{(2, 2), (3, 2), (4, 2), (2, 4)\}$

(C) $R_3 = \{(3, 3), (4, 3), (5, 4), (3, 4)$

(D) none of these

Q.15

Let A and B be two sets containing 2 elements and 4 elements respectively. The number of subsets of $A \times B$ having 3 or more elements is

(A) 211

(B) 256

(C) 220

(D) 219

Q.16

Let Q be the set of all rational numbers and R be the relation defined as $R = \{(x, y) : 1 + xy > 0, x, y \in Q\}$ Then relation R is

(A) symmetric and transitive

(B) reflexive and transitive

(C) an equivalence relation

(D) reflexive and symmetric

Q.17

Let $X = \{1, 2, 3, 4, 5\}$. The number of different ordered pairs (Y, Z) that can be formed such that $Y \subseteq X, Z \subseteq X$ and $Y \cap Z$ is empty is

(A) 2^5

(B) 5^3

(C) 5^2

(D) 3^5

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Q.18

If $A\Delta B = A \cup B$ then

(A) $A = B$

(B) $A \cap B = \phi$

(C) $A\Delta B = \phi$

(D) $A\Delta B = A \sim B$

Q.19

If $n(A) = n(B) = 4$ then number of bijections from A to B is

(A) 6

(B) 24

(C) 12

(D) 18

Q.20

If R and S are two symmetric relations then

(A) R o S is a symmetric relation

(B) $S \circ R$ is a symmetric relation

(C) R o S^{-1} is a symmetric relation

(D) RoS is a symmetric relation if and only if $RoS =$ SoR.

Q.21

Let $X = \{1, 2, 3, 4\}$. The number of equivalence relations that can be defined on X is

Q.22

From 50 students taking examinations in Mathematics, Physics and Chemistry, 37 passed Mathematics, 24 Physics and 43 Chemistry. At most 19 passed Mathematics and Physics, at most 29 Mathematics and Chemistry and at most 20 Physics and Chemistry. The largest possible number that could have passed all three exams is

Q.23

A relation on the set $A = \{x : |x| < 3, x \in \mathbf{Z}\}$, where $\mathbf{Z}$ is the set of integers is defined by $R = \{(x, y) : y = |x|, \quad x \neq -1\}$. Then the number of elements in the power set of R is

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Q.24

Suppose $A_1, A_2, \ldots, A_{30}$ are thirty sets each having 5 elements and $B_1, B_2 \ldots, B_n$ are n sets each with 3 elements, let $\bigcup_{i=1}^{30} A_i = \bigcup_{j=1}^{n} B_j = S$ and each element of S belongs to exactly 10 of the A_i 's and exactly 9 of the B_j 's. Then n is equal to

Q.25

If $A = \{1, 2, 3, 4\}, B = \{3, 4, 5\}$ then the number of elements in $(A \cup B) \times (A \cap B) \times (A \Delta B)$ is

Answers & Solutions

Q.1 Answer:

9

Solution:

Let S = the set of all members owning sailboats and P = the set of all members owning powerboats

$$n(S \cup P) = n(S) + n(P) - n(S \cap P)$$
$$82 = 53 + 38 - n(S \cap P)$$
$$\Rightarrow \quad n(S \cap P) = 91 - 82 = 9$$

Q.2 Answer:

transitive

Solution:

$2 \not> 2$ so $>$ is not reflexive, $3 > 2$ but $2 < 3$ so $(3, 2) \in R$ but $(2, 3) \notin R$. Thus R is not symmetric. If $(a, b) \in R$ and $(b, c) \in R$ then $a > b, b > c \Rightarrow a > c$ so $(a, c) \in R$ is not an equivalence relation.

Q.3 Answer:

9

Solution:

$A = \{D, E, N, S, T, U]$
$B = \{E, G, O, P, R, S\}$
$C = \{C, E, G, N, O, R, T, U\}$
So $A \cup (B \cap C)$
$= \{D, E, N, S, T, U\} \cup \{E, G, O, R\}$
$= \{D, E, G, O, N, R, S, T, U\}$
and $n[(A \cup (B \cap C)] = 9$

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Q.4 Answer:

$34 \in X \cap Y$

Solution:

If $n^3+2n+1=3n^2+7 \Rightarrow \quad n^3-3n^2+2n-6=0 \Rightarrow \quad (n-3)\left(n^2+2\right)=0$

$\Rightarrow \quad n=3$ **as** $n \in \mathbf{N}$

$\Rightarrow \quad \Rightarrow \quad n=3$ as $n \in \mathbf{N}$

So, $\quad x=3 \times 3^2+7=34 \in X \cap Y.$

In (a) and (b) $x \neq 34$**, for any** $n \in \mathbf{N}$**.**

Q.5 Answer:

symmetric

Solution:

For $a \in N$**,** a **is a factor of** a **so** R **is reflexive. If** a **is factor** b **and** b **is factor of** c **then** a **is factor of** c **so** R **is transitive. If** a **is factor of** b **and** b **is factor of** a **then** $a=b$ **so** R **is anti symmetric, 2 is factor of 4 but 4 is not a factor of** 2**.**

Q.6 Answer:

none of these

Solution:

Let $A=\{1,2,3,4,5\}, B=\{6,7,8,9\}$ **and** $C=\{11,12,13\}$ **which satisfy the given conditions but none of (a), (b) or (c).**

Q.7 Answer:

6,3

Solution:

According to the given condition, we have $2^m=2^n+56 \Rightarrow 2^{m-3}-2^{n-3}=7 \Rightarrow$
$2^{n-3}\left(2^{m-n}-1\right)=7$**. Since 7 is a prime number so we must have** $n-3=0$ **(clearly** $m \neq n$**).**
Thus $n=3$**. Therefore,** $2^m=2^3+56=64=2^6 \Rightarrow m=6$

Q.8 Answer:

reflexive, symmetric and not transitive

Solution:

$(x, x) \in \mathbf{R} \quad \forall x \in W$ **as all letters in both are common. If** $(x, y) \in \mathbf{R}$ **then** x **and** y **have a letter in common** $\Rightarrow (y, x) \in \mathbf{R}$

Next, let $x=$ **fix,** $y=$ six **and** $z=$ **son then** $(x, y) \in \mathbf{R}\ (y, z) \in \mathbf{R}$ **but** $(x, z) \notin \mathbf{R}$

So $\mathbf{R}$ **is reflexive, symmetric but not transitive**

Q.9 Answer:

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R is not reflexive, symmetric and not transitive

Solution:

$R = \{(1,4),(4,1),(2,3),(3,2)\}$, so R is not reflexive as $(1,1) \notin R$. R is symmetric by definition and R is not transitive as $(1,4) \in R, (4,1) \in R$ but $(1,1) \notin R$

Q.10 Answer:

$34 \leq x \leq 64$

Solution:

Let the total number of workers be 100 . A, the set of workers who like oranges and B, the set of workers who like apples.

So $\quad n(A) = 70, n(B) = 64, n(A \cap B) = x$

Also $n(A \cup B) \leq 100.$

$\Rightarrow n(A) + n(B) - n(A \cap B) \leq 100$

$\Rightarrow 70 + 64 - x \leq 100 \Rightarrow x \geq 34$

Since $n(A \cap B) \leq n(B) \Rightarrow x \leq 64$

Hence $34 \leq x \leq 64.$

Q.11 Answer:

B

Solution:

It can be shown by induction that $9 \mid 4^n - 3n - 1$ for every $n \in \mathbf{N}$. Thus $A \subseteq B$. Clearly $A \neq B$ as $27 \in B$ but $27 \notin A$. Thus $A \cup B = B$.

Q.12 Answer:

8

Solution:

$$
\begin{aligned}
& n\left(A \cup B'\right) = n(A) + n\left(B'\right) - n\left(A \cap B'\right) \\
& = n(A) + n(U) - n(B) - (n(A) - n(A \cap B)) \\
& = n(U) - n(B) + n(A \cap B) \\
& = 12 - 6 + 2 = 8
\end{aligned}
$$

Q.13 Answer:

transitive

Solution:

If $a = -1$ than a is not related to a. So R is not reflexive. If $a = -1, b = 2$ then $|a| \leq b$ but $|b| > a$, so R is not symmetric. If $|a| \leq b$ and $|b| \leq c$ then $|a| \leq b \leq |b| \leq c$ so $|a| \leq c$, hence a is related to c. Thus R is transitive. If $|a| \leq b$, and $|b| \leq a$ then $a \leq |a| \leq b \leq |b| \leq a$. So $a = b$, R is anti symmetric.

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Q.14 Answer:

none of these

Solution:

We have $\mathrm{R}=\left\{(\mathrm{x}, \mathrm{y}):\left|\mathrm{x}^2-\mathrm{y}^2\right|<16\right\}$

Take $\mathrm{x}=1$**, we have**

$\left|x^2-y^2\right|<16 \Rightarrow\left|1-y^2\right|<16$

$\Rightarrow\left|y^2-1\right|<16 \Rightarrow y=1,2,3,4$

Take $\mathrm{x}=2$**, we have**

$\left|x^2-y^2\right|<16 \Rightarrow\left|4-y^2\right|<16$

$\Rightarrow\left|y^2-4\right|<16 \Rightarrow y=1,2,3,4$

Take $\mathrm{x}=3$**, we have**

$x^2-y^2\left|<16 \Rightarrow 9-y^2\right|<16$

$\Rightarrow\left|y^2-9<16 \Rightarrow y=1,2,3,4\right.$

Take $x=4$**, we have**

$\left|x^2-y^2\right|<16 \Rightarrow\left|16-y^2\right|<16$

$\Rightarrow\left|y^2-16\right|<16 \Rightarrow y=1,2,3,4,5$

Take $\mathrm{x}=5$

$\mathrm{x}^2-\mathrm{y}^2|<16 \Rightarrow|25-\mathrm{y}^2|<16$

$\Rightarrow\left|\mathrm{y}^2-25\right|<16 \Rightarrow \mathrm{y}=4,5$

$\therefore \mathrm{R}=$

$\{(1,1),(1,2),(1,3),(1,4),(2,1),(2,2),(2,3),(2,4),(3,1),(3,2),(3,3),(3,4),(4,1)$

$\therefore$ **The correct answer is** D.

Q.15 Answer:

219

Solution:

$A \times B$ **contain** $(2)(4)=8$ **elements.**

The number of subsets of $\mathrm{A} \times \mathrm{B}$ **having 3 or more elements**

$=$ **Number of subsets of** $\mathrm{A} \times \mathrm{B}-[$ **number of subsets with at most 2 elements** $]$

$=2^8-\left({}^8C_1+{}^8C_1+{}^8C_2\right)=219$

Q.16 Answer:

reflexive and symmetric

Solution:

Since $1+x^2>0$ **for all** $x \in Q$**, so** R **is reflexive. If**

$(x, y) \in R \quad \Rightarrow \quad 1+x y>0$

$\Rightarrow \quad 1+y x>0 \Rightarrow(y, x) \in R$

Thus R **is symmetric. Take** $x=2, y=-\frac{1}{4}, z=-1$

$1+x y=1-\frac{1}{2}=\frac{1}{2}>0 \quad$ **so** $(x, y) \in R$

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$1 + yz = 1 + \frac{1}{4} = \frac{5}{4} > 0$ **so** $(y, z) \in R$

$1 + xz = 1 - 2 = -1 \Rightarrow (x, z) \notin R$. **Hence** R **is not transitive.**

Q.17 Answer:

3^5

Solution:

For each $x \in X$**, we have the following four choices:**

(a) $x \in Y$ **and** $x \in Z$

(b) $x \notin Y$ **and** $x \in Z$

(c) $x \in Y$ **and** $x \notin \mathrm{Z}$

(d) $x \notin Y$ **and** $x \notin Z$

As we do not want $x \in Y \cap Z$**, we are left with three choices for each** $x \in X$**.**

Thus, the total number of ways in which Y **and** Z **can be chosen is** 3^5**.**

Q.18 Answer:

$A \cap B = \phi$

Solution:

$A \cup B = A \Delta B \cup (A \cap B)$

So $A \cup B = A \Delta B \Rightarrow A \cap B = \phi$**.**

Q.19 Answer:

24

Solution:

As we have learned

Number of Bijective Function -

If A& B **are Bijective then** $n(A) = n(B) = m$ **(Say)**

So number of Bijective functions= $m!$

number of bijective function $= 4!$

Q.20 Answer:

RoS is a symmetric relation if and only if $RoS =$ SoR.

Solution:

Since R **and** S **are symmetric relations so** $R^{-1} = R$ **and** $S^{-1} = S$**.**

But $(R \circ S)^{-1} = S^{-1} o R^{-1} = SoR$**.**

Thus RoS **is symmetric if and only if** RoS $=$ **SoR**

Q.21 Answer:

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Solution:

The number of equivalence relations

$$B_k = \sum_{n=0}^{k-1} \left(\begin{array}{c} k-1 \\ n \end{array} \right) B_n; \quad B_0 = 1, B_1 = 1, B_2 = 2, B_3 = 5$$

$$B_4 = \left(\begin{array}{c} 3 \\ 0 \end{array} \right) B_0 + \left(\begin{array}{c} 3 \\ 1 \end{array} \right) B_1 + \left(\begin{array}{c} 3 \\ 2 \end{array} \right) B_2 + \left(\begin{array}{c} 3 \\ 3 \end{array} \right) B_3$$

$$= 1 + 3 + 3 \times 2 + 1 \times 5 = 15$$

Q.22 Answer:

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Solution:

The given conditions can be expressed as

$$n(M \cup P \cup C) = 50, n(M) = 37, n(P) = 24,$$

$$n(C) = 43, n(M \cap P) \leq 19,$$

$$n(M \cap C) \leq 29 \text{ and } n(P \cap C) \leq 20,$$

$$n(M \cup P \cup C) = n(M) + n(P) + n(C) - n(M \cap P) - n(M \cap C) - n(P \cap C) + n(M \cap P \cap C)$$

$$\Rightarrow \quad 50 = 37 + 24 + 43 - n(M \cap P) - n(P \cap C) - n(M \cap C) + n(M \cap P \cap C)$$

$$\Rightarrow \quad n(M \cap P \cap C) \leq n(M \cap P) + n(M \cap C) + n(P \cap C) - 54$$

Therefore, the number of students that could have passed all three exams is at most $19 + 29 + 20 - 54 = 14$

Q.23 Answer:

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Solution:

$$A = \{-2, -1, 0, 1, 2\}$$

$$R = \{(-2, 2), (0, 0), (1, 1), (2, 2)\}$$

As R **has four elements, the power set of** R **contains 16 elements.**

Q.24 Answer:

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Solution:

$S = \bigcup_{i=1}^{30} A_i$, **so** $n(S) = \frac{1}{10}(5 \times 30) = 15$ **(since element in the union** S **belongs to exactly 10 of the** A_i **'s). Again** $S = \bigcup_{i=1}^{n} B_i$ **so**

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$n(S) = 1/9(3 \times n) = n/3 = 15 \Rightarrow n = 45$

Q.25 Answer:

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Solution:

$$\begin{aligned} A \cup B &= \{1,2,3,4,5\}. n(A \cup B) = 5 \\ A \cap B &= \{3,4\}, n(A \cap B) = 2 \\ A \Delta B &= (A \sim B) \cup (B \sim A) = \{1,2\} \cup \{5\} \\ &= \{1,2,5\} \\ n(A \Delta B) &= 3. \text{ Hence } n((A \cup B) \times (A \cap B) \times (A \Delta B)\} \\ &= 5 \times 2 \times 3 = 30 \end{aligned}$$

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